CS 70 离散数学 和 概率论
笔记 17
春季 2026 Course Notes

# Functions of Random Variables

In the previous 笔记, we considered simple 线性的 functions (sums, differences 和 常数 multiples) of randomvariables. Inthisnote,wewillconsidermoregeneralfunctionsofarandomvariable. Formally, 令 $X$ be a random 变量 on a 样本空间 with 概率 分布 $P_X$. 那么, 对于 a 函数 $f(\cdot)$ on the 值域 of $X$, $f(X)$ is also a random 变量 on the same 样本空间. Denoting $Y = f(X)$, wehave $Y(\omega) = f(X(\omega))$ for $\omega \in$ .

Intermsofsamplespaces,theevent "$Y = y$" isequivalenttotheevent "$X \in f^{-1}(y)$", 其中 $f^{-1}(y)$ denotes thepreimage $\{x \mid f(x) = y\}$. (When $f(\cdot)$ isone-to-one,itisabitsimpler,i.e., "$X = x$" isthesameeventas "$Y = f(x)$".)

Withthisview,thedistributionof $Y = f(X)$, $P_Y$, canbederivedfromthedistributionof $X$ asfollows:

$$P_Y[Y = y] = \sum_{x: f(x) = y} P_X[X = x]. \quad (1)$$

Fromthis,weobservethat

$$E[Y] = \sum_y y\, P_Y[Y = y] = \sum_y \sum_{x: f(x) = y} y\, P_X[X = x] = \sum_x f(x) P_X[X = x].$$

Thefirstequalityisbythedefinitionofexpectation,thesecondis(1),andthelastcomesbyobservingthat eachvalueofxisincludedexactlyonceintheinnersummation.

因此,wehavethefollowingidentity:

$$E[f(X)] = \sum_x f(x) P_X[X = x].$$

Inwords: tocomputetheexpectationof $f(X)$,wetakethesameweightedaverageasintheusualexpectation of $X$, 但是 replace the value $x$ of $X$ by $f(x)$. (This identity is sometimes called the Law of the Unconscious Statistician(LOTUS)asitisusefulandusedwithoutmuchthoughtattimes.)

Anexamplethatwewilluseextensivelyisthefollowing.

Foranyr.v.X,wehave $E\left[X^2\right] = \sum_x x^2 P_X[X = x]$.

Warning: 笔记 that $E\left[X^2\right]$ is 非 the same as $(E[X])^2$. Similarly, in general, $E[f(X)]$ is 非 the same as $f(E[X])$, 除非 $f$ isoftheform $f(X) = aX + b$ where $a$ and $b$ areconstants.

# Random Variables: 方差 和 Covariance

Wehaveseeninthepreviousnotethatifwetakeabiasedcointhatcomesupheadswithprobability $p$ and tossit $n$ times,thentheexpectednumberofheadsis $np$. Whatthismeansisthatifwerepeattheexperiment multipletimes,whereineachexperimentwetossthecoin $n$ times,thenonaveragewegetnpheads. Butin any single experiment, the number of heads observed can be any value between 0 和 n. What can we say about how far off we are from the expected value? 即, what is the typical deviation of the number of headsfromnp?



## 1 条件的 期望 和 the Law of Total 期望

LetX andY berandomvariablesdefinedonthesameprobabilityspace,takingvaluesinA andB,respec- tively. 给定 their joint 分布 $\{((a,b),P[X=a,Y=b]):a\in A,\ b\in B\}$, we 定义 the 条件的 probabilityofX =agivenY =b(forP[Y =b]>0)as

$$P[X=a\mid Y=b]=\frac{P[X=a,Y=b]}{P[Y=b]}.$$

Thisleadstothenotionofconditionalexpectation:

定义 17.1 (条件的 期望). 对于 any $b\in B$ with $P[Y=b]>0$, the 条件的 期望 ofX givenY =bis

$$E[X\mid Y=b]=\sum_{a\in A} a\cdot P[X=a\mid Y=b].$$

因此, $E[X\mid Y=b]$ is simply the expected value of X conditioned on the eventY =b. 对于 each fixed b, it satisfiestheusualpropertiesofexpectation,suchaslinearity.

Nowdefinetherandomvariable

$$f(Y)=E[X\mid Y].$$

Thatis, f(Y)takesthevalueE[X |Y =b]whenY =b. 因为 f(Y)isafunctionoftherandomvariableY,it isitselfarandomvariabledefinedonthesameprobabilityspace. Itsexpectationisgivenby

$$E[f(Y)]=\sum_{b\in B} f(b)P[Y=b]=\sum_{b\in B} E[X\mid Y=b]P[Y=b]. \quad (2)$$

ThefollowingtheoremshowsthatthisquantityequalsE[X].

Theorem17.1(LawofTotalExpectation). If$E[|X|]<\infty$,那么

$$E[X]=E[E[X\mid Y]].$$

(ThisisalsoknownastheLawofIteratedExpectationortheTowerRule.)

证明. Using(2),wehave

$$\begin{aligned}E[E[X\mid Y]]&=\sum_{b\in B}E[X\mid Y=b]P[Y=b]\\&=\sum_{b\in B}\sum_{a\in A}aP[X=a\mid Y=b]P[Y=b]\\&=\sum_{a\in A}a\sum_{b\in B}P[X=a\mid Y=b]P[Y=b]\\&=\sum_{a\in A}aP[X=a]\\&=E[X].\end{aligned}$$

InthethirdlineweinterchangetheorderofsummationandthefourthlinefollowsfromtheTotalProbability Rule.

Asaresult,Theorem17.1impliesthatwecancomputeE[X]as

$$E[X]=\sum_{b\in B}E[X\mid Y=b]P[Y=b], \quad (3)$$

whichisoftenusefulbecauseconditioningonY =bcansimplifythecomputation.



## 2 Random Walk

令 us 考虑 a simpler setting 即 等价 to tossing a fair coin n times, 但是 is easier to picture.

time step, the particle moves either one step to the right 或
one step to the left with equal 概率 (this

kindofrandomwalkiscalledsymmetric),andthemoveateachtimestepisindependentofallothermoves.
We think of these random moves as taking place according to
whether a fair coin comes up heads 或 tails.
The expected position of the particle after n moves is back
at 0, 但是 how far from 0 should we typically
expecttheparticletoendup?
Denotingaright-moveby+1andaleft-moveby  1,
wecandescribetheprobabilityspacehereastheset

ofallsequencesoflengthnoverthealphabet{±1},eachhavingequalprobability
1. Letther.v.S 表示
2n n

thepositionoftheparticle(relativetoourstartingpoint0)afternmoves.
因此,wecanwrite
S =X +X +・・・+X , (4)
n 1 2 n
whereX =+1iftheithmoveistotherightandX
=  1ifthemoveistotheleft.
i i
TheexpectationofS canbeeasilycomputedasfollows.
SinceE[X]=(1×1)+(1×(  1))=0,applying
n i 2 2
linearityofexpectationimmediatelygivesE[S
n
]=Σ n
i=1
E[X
i
]=0. Butofcoursethisisnotveryinformative,

andisduetothefactthatpositiveandnegativedeviationsfrom0cancelout.
Whatwearereallyaskingis: Whatistheexpectedvalueof|S
|,thedistanceoftheparticlefrom0?
n
Ratherthanconsiderther.v.|S
|,whichisalittledifficulttoworkwithduetotheabsolutevalueoperator,we
n
will instead look at the r.v. S2. Notice that this also has
the effect of making all deviations from 0 正的,
n
所以 it should also give a good measure of the distance from
0. 然而, because it is the squared distance,
wewillneedtotakeasquarerootattheend.

Wewillnowshowthattheexpectedsquareddistanceafternstepsisequalton:
Proposition17.1. FortherandomvariableS
definedin(4),wehaveE(cid:2) S2(cid:3)
=n.
n n
证明. Weusetheexpressionfrom(4)andexpandthesquare:
(cid:34) (cid:35)
n n
E(cid:2) S2(cid:3) =E(cid:2) (X +X +・・・+X )2(cid:3) =E
ΣX2+2ΣXX = ΣE(cid:2) X2(cid:3) +2ΣE[XX ]. (5)

n 1 2 n i i j i i j
i=1 i<j i=1 i<j
In the last equality 我们有 used linearity of 期望. To proceed, we need to 计算
E(cid:2) X2(cid:3)
和
i
E[XX ] 对于 i = j. 因为 X can take on only values ±1, clearly X2 =1 always, 所以 E(cid:2) X2(cid:3) =1. To 计算
i j i i i
E[XX ]fori = j,noteXX =+1whenX =X =+1orX =X = 1,andotherwiseXX = 1. 因此,
i j i j i j i j i j
P[XX =1]=P[(X =X =+1)∨(X =X = 1)]
i j i j i j
=P[X =X =+1]+P[X =X = 1]
i j i j
=P[X =+1]×P[X =+1]+P[X = 1]×P[X = 1]
i j i j
1 1 1
= + = ,
4 4 2
wherethesecondequalityfollowsfromthefactthattheeventsX =X =+1andX =X = 1aremutually
i j i j
exclusive, 当...时 the third equality follows from the independence of the events X =+1 和 X =+1, 和
i j
likewise 对于 the events X = 1 和 X = 1. 因为 the only other possible value 对于 XX is 1, we must
i j i j


also have P[XX = 1] = 1, 和 因此 E[XX ] = 0. [笔记: Later we will see a much simpler way to
i j 2 i j
computeE[XX ]usingthefactthatX andX areindependent.]
i j i j
Finally, plugging E(cid:2) X
i
2(cid:3) =1 和 E[X
i
X
j
]=0, 对于 i = j, into (5) gives E(cid:2) S
n
2(cid:3) =∑ n
i=1
1+2∑
i<j
0=n, as
claimed.
所以, 对于 the 对称的 random walk 例子, we see that theexpectedsquareddistancefrom0isn. One
√
interpretationofthisisthatwemightexpecttobeadistanceofabout nawayfrom0afternsteps. 然而,
wehavetobecarefulhere: wecannotsimplyarguethatE[|S |]=
(cid:112)E[S2]= √ n. (Whynot?) Wewillsee
n n

later in the course how to make precise deductions about |S
| from knowledge of
E(cid:2) S2(cid:3)
. 对于 the moment,
n n
然而,letusagreetoviewE(cid:2) S2(cid:3)
asanintuitivemeasureof“spread”ofther.v.S .
n n
对于 a more general r.v. X with 期望 E[X]=μ, what we are really
interested in is E(cid:2) (X μ)2(cid:3) , the
expectedsquareddistancefromthemean.
Inoursymmetricrandomwalkexample,wehaveE[S ]=μ =0,
n
soE(cid:2)
(S
μ)2(cid:3) justreducestoE(cid:2) S2(cid:3)
.
n n
Definition17.2(方差). Forar.v.X
withexpectationE[X]=μ,thevarianceofX isdefinedtobe
Var(X)=E(cid:2) (X μ)2(cid:3)
.
(cid:112)
Thesquarerootσ(X):= Var(X)iscalledthestandarddeviationofX.

Thepointoftakingthesquarerootofvarianceistoputthestandarddeviation“onthesamescale”asther.v.
itself. 因为 the 方差 和 标准差 differ just by a square, it really
doesn’t matter 其中 one
we choose to work with as we can always 计算 one from the
other. We shall usually use the 方差.

Fortherandomwalkexampleabove,Proposition17.1impliesthatVar(S
)=n,andthestandarddeviation
√ n
isσ(S )= n.
n

Thefollowingobservationprovidesaslightlydifferentwaytocomputethevariance,whichsometimesturns
outtobesimpler.
Theorem17.2. Forar.v.X
withexpectationE[X]=μ,wehaveVar(X)=E(cid:2) X2(cid:3) μ2.
证明. Fromthedefinitionofvariance,wehave
Var(X)=E(cid:2) (X μ)2(cid:3) =E(cid:2) X2 2μX+μ2(cid:3)
=E(cid:2) X2(cid:3) 2μE[X]+μ2=E(cid:2) X2(cid:3) μ2.
In the third equality, we used linearity of 期望; 笔记 that, 因为
μ = E[X] is a 常数, E[μX] =
μE[X]=μ2 andE(cid:2) μ2(cid:3) =μ2.
Another important property that will come in handy is the
following: 对于 any random 变量 X 和 con-
stantc,wehave
Var(cX)=c2Var(X). (6)
Verifyingthisisstraightforwardandleftasanexercise.
3 方差 Computation
Letusseesomeexamplesofvariancecalculations.


1. Fairdie. LetX bethescoreontherollofasinglefairdie.
RecallfromthepreviousnotethatE[X]= 7.
SowejustneedtocomputeE(cid:2) X2(cid:3)
,whichisaroutinecalculation:

2
E(cid:2) X2(cid:3)
=
1(cid:0) 12+22+32+42+52+62(cid:1)
=
91
.
6 6
因此,fromTheorem17.2,
Var(X)=E(cid:2) X2(cid:3) (E[X])2= 91 49 = 35 .
6 4 12
2. Uniformdistribution. Moregenerally,supposeX
isauniformrandomvariableontheset{1,...,n},
written X 均匀的{1,...,n}, 所以 X takes on values 1,...,n with
equal 概率 1. The mean,
n
varianceandstandarddeviationofX aregivenby:
(cid:114)
n+1 n2 1 n2 1
E[X]= , Var(X)= , σ(X)= . (7)
2 12 12
Youshouldverifytheseasanexercise. [Recallthat∑ n i= n(n+1)
和∑ n i2= n(n+1)(2n+1) .]
i=1 2 i=1 6
3. Fixed points of permutations. 令 X be the number of fixed
points in a random permutation of n
n
items(i.e.,inthehomeworkpermutationexample,X
isthenumberofstudentsinaclassofsizenwho
n
receive their own 作业 after shuffling). We saw in the
previous 笔记 that E[X ]=1, regardless
n
of n. To 计算
E(cid:2) X2(cid:3)
, write X =I +I +···+I , 其中 I =1 如果 i is a fixed point, 和 I
=0
n n 1 2 n i i
otherwise. Thenasusualwehave
n
E(cid:2) X2(cid:3) = ∑E(cid:2) I2(cid:3) +2∑E[II ]. (8)
n i i j
i=1 i<j
因为 I is an indicator r.v., 我们有 that E(cid:2) I2(cid:3) =P[I
=1]= 1. 对于 i< j, 因为 both I 和 I are
i i i n i j
indicators,wecancomputeE[II ]asfollows:
i j
1
E[II ]=P[II =1]=P[I =1∧I =1]=P[bothiand j arefixedpoints]= .
i j i j i j
n(n 1)
Makesurethatyouunderstandthelaststephere!
Pluggingthisinto(8)weget
E(cid:2) X2(cid:3)
= ∑
n 1
+2∑

1
=
(cid:18)
n×
1 (cid:19)
+
(cid:20)
2
(cid:18) n (cid:19)
×
1 (cid:21)
=1+1=2.
n n n(n 1) n 2 n(n 1)
i=1 i<j
因此, Var(X )=E(cid:2) X2(cid:3) (E[X ])2 =2 1=1. 即, the 方差 和
the mean are both equal
n n n
to 1. Like the mean, the 方差 is also 独立的 of n. Intuitively at
least, this means that it is

unlikelythattherewillbemorethanasmallnumberoffixedpointsevenwhenthenumberofitems,
n,isverylarge.
4 Sum of 独立的 Random Variables
One of the most important 和 useful facts about 方差 is that 如果
a random 变量 X is the sum of
independentrandomvariablesX=X +・・・+X
,thenitsvarianceisthesumofthevariancesoftheindividual
1 n
r.v.’s. Inparticular,iftheindividualr.v.’sX
areidenticallydistributed(i.e.,theyhavethesamedistribution),
i √
那么 Var(X)=∑ n
i=1
Var(X
i
)=n・Var(X
1
). This means that the 标准差 is σ(X)= n・σ(X
1
).


Notethatbycontrast,theexpectedvalueisE[X]=n・E[X ].
Intuitivelythismeansthatwhereastheaverage
1 √
value of X grows proportionally to n, the spread of the 分布
grows proportionally to n, 其中 is
muchsmallerthann. Inotherwords,thedistributionofX
tendstoconcentratearounditsmean.
Letusnowformalizetheseideas.
First,wehavethefollowingresultwhichstatesthattheexpectedvalueof

theproductoftwoindependentrandomvariablesisequaltotheproductoftheirexpectedvalues.
Theorem17.3.
ForindependentrandomvariablesX,Y,wehaveE[XY]=E[X]E[Y].
证明. Wehave
E[XY]=∑∑ab×P[X =a,Y =b]
a b
=∑∑ab×P[X =a]×P[Y =b]
a b

(cid:32) (cid:33)
(cid:18) (cid:19)
= ∑a×P[X =a] × ∑b×P[Y =b]
a b
=E[X]×E[Y],
asrequired.
Inthesecondlineherewemadecrucialuseofindependence.
We now use the above 定理 to conclude the nice property that
the 方差 of the sum of 独立的
randomvariablesisequaltothesumoftheirvariances.
Theorem17.4. ForindependentrandomvariablesX,Y,wehave
Var(X+Y)=Var(X)+Var(Y).
证明.
FromthealternativeformulaforvarianceinTheorem17.2andlinearityofexpectation,wehave
Var(X+Y)=E(cid:2) (X+Y)2(cid:3)   (E[X+Y])2
=E(cid:2) X2(cid:3) +E(cid:2) Y2(cid:3) +2E[XY]  (E[X]+E[Y])2
=(E(cid:2) X2(cid:3)   E[X]2)+(E(cid:2) Y2(cid:3)
  E[Y]2)+2(E[XY]  E[X]E[Y])
=Var(X)+Var(Y)+2(E[XY]  E[X]E[Y]).
SinceX,Y
areindependent,Theorem17.3impliesthatthefinalterminthisexpressioniszero.
It is very important to remember that neither of the above
two results is 真 in general when X,Y are 非
独立的. Asasimpleexample,notethatevenfora{0,1}-valuedr.v.X
withP[X =1]= p,E(cid:2) X2(cid:3) = p
isnotequaltoE[X]2= p2 (becauseofcourseX andX
arenotindependent!). Thisisincontrasttolinearity
of 期望, 其中 we saw that the 期望 of a sum of r.v.’s is the sum
of the expectations of the

individualr.v.’s,regardlessofwhetherornotther.v.’sareindependent.
例子
令 us return to our motivating 例子 of a 序列 of n coin tosses. 令
X 表示 the number of
n
Headsinntossesofabiasedcoinwithheadsprobability p(i.e.,X
  二项的(n,p)). Asusual,wewrite
n
X =I +I +• • • +I ,whereI =1iftheithtossisH,andI =0otherwise.
n 1 2 n i i


We already know E[X
n
]=∑ n
i=1
E[I
i
]=np. We can 计算 Var(I
i
)=E(cid:2) I
i
2(cid:3)   E[I
i
]2 = p  p2 = p(1  p).
SincetheI
i
’sareindependent,byTheorem17.4wegetVar(X
n
)=∑ n

$i=1$
Var(I
i
)=np(1 p).
As an 例子, 对于 a fair coin (p = 1) the expected number of
Heads in n tosses is n, 和 the standard
√ 2 2
deviationis (cid:112)n = n .
Notethatsincethemaximumnumberof Headsisn, thestandarddeviationismuch
4 2
less than this maximum number 对于 large n. This is in
contrast to the previous 例子 of the uniformly
(cid:113)

distributedrandomvariablein(7), wherethestandarddeviation σ (X)=
n2 1 ≈ √n (forlargen)isofthe
12 12
same order as the largest value, n. In this sense, the
spread of a binomially distributed r.v. is much smaller
thanthatofauniformlydistributedr.v.

## 5 Covariance 和 Correlation

The expression E[XY] E[X]E[Y] in the 证明 of 定理 17.4 is a
measure of association between X,Y,
andiscalledthecovariance:
定义 17.3 (Covariance). The covariance of random variables X
andY, denoted cov(X,Y), is defined
as
cov(X,Y)=E[(X μ )(Y μ )]=E[XY] E[X]E[Y],
X Y
whereμ =E[X]和 μ =E[Y].
X Y
练习:
Checkthattheabovetwoexpressionsforcov(X,Y)areindeedequal.
Remarks. Wenotesomeimportantfactsaboutcovariance.
1. IfX,Y areindependent, thencov(X,Y)=0.
然而, theconverseisnottrue.
2. cov(X,X)=Var(X).
3.
Covarianceisbilinear;i.e., foranycollectionofrandomvariables{X
,...,X }, {Y ,...,Y }andfixed
1 n 1 m
constants{a ...,a }, {b ,...,b },
1 n 1 m
cov(Σ n i=1 a i X i , Σ m j=1 b j Y j )=Σ n i=1 Σ m j=1 a i b
j cov(X i ,Y j ).
ForgeneralrandomvariablesX andY,
Var(X+Y)=Var(X)+Var(Y)+2cov(X,Y).

Whilethesignofcov(X,Y)(positiveornegative)isinformativeofhowX
andY areassociated, itsmagnitude
isdifficulttointerpret.
Astatisticthatiseasiertointerpretiscorrelation:
Definition17.4(Correlation). SupposeX andY
arerandomvariableswith σ (X)>0and σ (Y)>0. 那么,
thecorrelationofX andY isdefinedas
cov(X,Y)
Corr(X,Y)= .
σ (X) σ (Y)

Correlation is more useful than covariance because the
former always ranges between 1 和 +1, as the
followingtheoremshows:

Theorem17.5. ForanypairofrandomvariablesX andY
withσ(X)>0andσ(Y)>0,
1≤Corr(X,Y)≤+1.



证明. 令 E[X] = μ
X
和 E[Y] = μ
Y
, 和 定义 X(cid:101) = (X μ
X
)/σ(X) 和 Y(cid:101) = (Y μ
Y
)/σ(Y). 那么,
E[X(cid:101) 2]=E[Y(cid:101) 2]=1,所以
0≤E[(X(cid:101) Y(cid:101))2]=E[X(cid:101) 2]+E[Y(cid:101)
2] 2E[X(cid:101)Y(cid:101)]=2 2E[X(cid:101)Y(cid:101)]
0≤E[(X(cid:101)+Y(cid:101))2]=E[X(cid:101) 2]+E[Y(cid:101)
2]+2E[X(cid:101)Y(cid:101)]=2+2E[X(cid:101)Y(cid:101)],
whichimplies 1≤E[X(cid:101)Y(cid:101)]≤+1. Finally,notethat
cov(X,Y) E[(X μ )(Y μ )] (cid:104) (cid:105)
Corr(X,Y)= = X Y =E X(cid:101)Y(cid:101) .
σ(X)σ(Y) σ(X)σ(Y)
因此,wecandeducethat 1≤Corr(X,Y)≤+1.

NotethattheaboveproofshowsthatCorr(X,Y)=+1ifandonlyifE[(X(cid:101) Y(cid:101))2]=0,whichimpliesX(cid:101)=Y(cic
withprobability1.
Similarly,Corr(X,Y)= 1ifandonlyifE[(X(cid:101)+Y(cid:101))2]=0,whichimpliesX(cid:101)= Y(cid:101)
with
probability1.
IntermsoftheoriginalrandomvariablesX,Y,thismeansthefollowing:
ifCorr(X,Y)=±1,
thenthereexistconstantsaandbsuchthat,withprobability1,
Y =aX+b,
wherea>0ifCorr(X,Y)=+1anda<0ifCorr(X,Y)= 1.
(Thevaluesa,bdependonμ ,μ , σ(X), σ(Y);
X Y
inparticular,a=±σ(Y)
,wherethesigndependsonwhetherCorr(X,Y)=±1.) ThusX andY
differjustby
σ(X)
ascalingfactorandashift.

## 6 Exercises

1. ForanyrandomvariableX
andconstantc,showthatVar(cX)=c2Var(X).
2. ForX 均匀的{1,...,n},showthatE[X]andVar(X)areasshownin(7).